

Transpositions of a Universal Hyperbolic Law: Invariants, Rigidity, and Cross-Domain Predictions (P5)

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Abstract

We state a universal stencil for bounded scalars driven by additive quantities: in rapidity $r = \operatorname{artanh}(u)$, first-order responses are affine with slope $\lambda = \alpha_{\text{sys}}\kappa$. We formalize three falsifiable invariants and prove four rigidity results (Hilbert-projective uniqueness, vector-field uniqueness, rank-1 symmetric-domain structure, and information-monotonicity selecting log-odds). We give concise one-page transpositions (Ising, SDT/psychophysics, machine learning, population genetics, SIR epidemics, saturated control, and market microstructure), specify a non-commuting Möbius generator pair whose boundary limit set is fractal, and provide an engineering falsifier (FLOW-GATE) with explicit statistics. This paper references but does not revise P1–P4; it is the deployable stencil that connects the prior framework across domains.

1 Position of this paper

We build on:

- **P1** Universal Hyperbolic Information Geometry (axioms).
- **P2** Hyperbolic Structure of Bounded Dynamical Systems (rapidity/logistic).
- **P3** Einstein Addition from Associativity & Boundary-Fixing Möbius.
- **P4** The Murray Reality Equation (MRE) and protocols.

Here we: (i) state domain-agnostic invariants; (ii) give compact rigidity theorems with explicit hypotheses; (iii) provide a reusable transposition stencil plus worked micro-examples; (iv) formalize the multi-drive (non-commuting) fractal limit-set claim with concrete generators; (v) specify an engineering falsifier with power targets; (vi) derive a clean energy–rapidity inequality.

2 Notation and core stencil

Notation

All bounded observables are mapped into $u \in (-1, 1)$. Rapidity is $r(u) := \operatorname{artanh}(u)$; logistic coordinate is $u = \tanh r$. Additive drives are denoted Ψ (time, bits, steps, generations, opportunities, etc.). Device/domain calibration enters via $\alpha_{\text{sys}} > 0$. The universal scale is $\kappa \geq 0$. In multi-D, Einstein gyroaddition governs composition; its *gyration* encodes non-associativity.

Definition 1 (Transposition stencil). Given a bounded scalar $X \in (a, b)$:

- (i) Map $X \mapsto u \in (-1, 1)$ linearly.
- (ii) Define $r = \text{artanh}(u)$.
- (iii) Choose an additive drive Ψ .
- (iv) Test $r(\Psi) = r_0 + \lambda\Psi$; read off $\lambda = \alpha_{\text{sys}}\kappa$; curvature fingerprints interactions/gyrations.

Definition 2 (Rapidity Linearity Principle (RLP)). For any bounded scalar $u \in (-1, 1)$ driven by an additive Ψ ,

$$r(\Psi) = r_0 + \lambda\Psi + o(\Psi) \quad \text{with} \quad \lambda = \alpha_{\text{sys}} \kappa.$$

3 Three falsifiable invariants

Proposition 1 (Composition-law uniqueness (1-D)). Let $\oplus : (-1/\kappa, 1/\kappa)^2 \rightarrow (-1/\kappa, 1/\kappa)$ be associative, strictly increasing in each argument, with identity 0 and inverses, and assume each left-translation extends to a boundary-fixing Möbius map. Then

$$x \oplus y = \frac{x + y}{1 + \kappa^2 xy}, \quad r(x) = \kappa^{-1} \text{artanh}(\kappa x),$$

unique up to scale $\kappa > 0$.

Proposition 2 (RLP diagnostic). If $u = \tanh r$ with r the unique linearizer from Proposition 1, then to first order under any additive drive Ψ , $r(\Psi)$ is affine. Deviations are second order or encode interactions (multi-D gyrations).

Proposition 3 (Energy–rapidity inequality). For a two-outcome device at temperature T , the minimal average heat/work Q to shift the rapidity by Δr satisfies

$$Q \geq \frac{k_B T}{\chi} \Delta r,$$

for a device constant $\chi > 0$ determined by the baseline occupancy and actuation pathway. Equality is approached by quasi-static protocols that monotonically tilt log-odds.

Remark 1. Proposition 3 follows from information thermodynamics: changing the Bernoulli parameter from p to p' costs $Q \geq k_B T D_{\text{KL}}(p' \| p)$; near baseline, $D_{\text{KL}} \asymp \frac{1}{2} I_F(\Delta r)^2$, and along minimal-tilt schedules the incremental work per unit r is bounded below by $(k_B T / \chi)$; see Appendix A for an explicit derivation and the definition of χ .

4 Rigidity pack (explicit hypotheses)

R1. Projective–Hilbert uniqueness on the interval

Theorem 1 (Hilbert-projective rigidity). Let $\text{PSL}(2, \mathbb{R})$ act on $(-1, 1)$ via boundary-fixing Möbius maps. The Hilbert projective metric

$$d_H(x, y) = \left| \log \frac{(1+y)(1-x)}{(1-y)(1+x)} \right|$$

is (up to scale) the unique complete, cross-ratio derived metric invariant under the action. In this metric the coordinate $r = \text{artanh}(x)$ linearizes all hyperbolic one-parameter subgroups, i.e. translations in r .

Proof sketch. For an interval (rank-1 cone section), the Hilbert metric is precisely the log cross-ratio. A hyperbolic one-parameter subgroup is conjugate in $\text{PSL}(2, \mathbb{R})$ to $\text{diag}(e^t, e^{-t})$ acting projectively; its orbits are geodesics with constant speed in d_H . The change of variables $r = \frac{1}{2} \log \frac{1+x}{1-x} = \text{artanh } x$ rectifies the flow. Uniqueness follows from projective invariance and completeness. $\square$

R2. Vector-field uniqueness (no other ODEs)

Theorem 2 (Vector-field rigidity). *Let $V(x)\partial_x$ be C^1 , complete on $(-1, 1)$, strictly increasing ($V > 0$), vanishing only at ± 1 , and projectively covariant: for any boundary-fixing Möbius M , there exists $c(M) > 0$ s.t. $M_*V = c(M)V$. Then*

$$\dot{x} = \lambda(1 - x^2), \quad \dot{r} = \lambda,$$

for some $\lambda \in \mathbb{R}$.

Proof outline. Projective covariance forces V to transform as a weight- -1 density. Completeness and simple zeros at ± 1 forbid higher-order vanishing; invariance under $x \mapsto -x$ pins a quadratic factor. Up to scale, $V(x) = 1 - x^2$, which is exactly the pushforward of a constant field in r . $\square$

R3. Rank-1 symmetric-domain structure (multi-D)

Theorem 3 (Ball model and gyrostructure). *Assume a connected, bounded, homogeneous and isotropic state space in higher dimension. Then it is (up to biholomorphism) the rank-1 bounded symmetric domain (the open ball), with isometry group acting transitively. Composition is the Einstein gyroaddition; non-associativity is carried by Thomas-type gyrations.*

Idea. By Cartan/Koecher/Vinberg classification, rank-1 bounded symmetric domains are balls with automorphism group $SU(1, n)$ acting transitively; restriction to real slices yields the Einstein gyrovector structure. Gyrations arise from the holonomy of composing boosts. $\square$

R4. Information-monotonicity selects log-odds

Theorem 4 (DPI $\Rightarrow$ log-odds additivity). *Let $\{P_\theta\}$ be the Bernoulli family with canonical (natural) parameter $\theta = \log \frac{p}{1-p}$. Among statistics that are Markov-monotone (respect data-processing) and additive under independent composition, the unique (up to scale) choice is θ , i.e. $r = \frac{1}{2}\theta = \text{artanh}(2p - 1)$.*

Sketch. Chentsov's theorem fixes the Fisher metric as the unique monotone Riemannian metric; exponential families are linear in the natural parameter θ . For Bernoulli, the canonical sufficient statistic is the count and the additive parameter is θ , hence $r = \theta/2$. $\square$

5 Transposition recipe (drop-in)

Step 1: Identify a bounded observable $X \in (a, b)$. Map linearly to $u \in (-1, 1)$. **Step 2:** Set $r = \text{artanh}(u)$. **Step 3:** Choose an additive drive Ψ (time, bits, steps, generations, contacts). **Step 4:** Fit $r(\Psi) = r_0 + \lambda\Psi$; record λ and residual curvature.

6 One-page transpositions (predictions & falsifiers)

Each entry gives u , Ψ , the rapidity prediction, and a crisp falsifier.

Mean-field Ising. $u = m \in (-1, 1)$, $\Psi = \beta h$. Mean field: $m = \tanh(\beta J m + \beta h)$. For $J = 0$: $r = \text{artanh}(m) = \beta h$ (exact line; slope = 1). *Falsifier:* curvature with $J = 0$.

SDT / sequential evidence. u = decision bias; Ψ = accumulated evidence (bits). Prediction: $r(\Psi) = r_0 + \lambda\Psi$; shuffled evidence destroys slope. *Falsifier:* slope persists under shuffle.

ML learning curves. $u = 2s - 1$ for accuracy $s \in (0, 1)$; $\Psi = \text{tokens/steps}$. Logistic $s(t) = 1/(1 + e^{-(\lambda t + b)})$ gives $r(t) = \text{artanh}(2s - 1) = \frac{1}{2}(\lambda t + b)$ (exact). *Falsifier*: nonlinearity in r absent curriculum/regularization changes.

Population genetics (Wright–Fisher, weak selection). $u = 2p - 1$; $\Psi = t$ generations; small s : $\frac{d}{dt} \text{artanh}(2p - 1) \approx s$. *Falsifier*: persistent curvature without drift/epistasis.

SIR epidemics (bounded prevalence). $u = \text{logit}(i)$ remapped to $(-1, 1)$; $\Psi = \text{contact opportunities}$. In well-mixed regime and single phase, r is affine; network effects bend it. *Falsifier*: curvature where well-mixed holds.

Control with saturation. $u \in (-1, 1)$ actuator; $\Psi = \text{integrated effort}$. Prediction: r affine until hard limits; deviations diagnose constraints. *Falsifier*: sublinear r without constraints.

Market microstructure. $u = \text{trade imbalance}$; $\Psi = \text{signed order flow}$. Short horizon: r affine; liquidity regime shifts bend slope. *Falsifier*: no horizon with linear r .

7 Non-commuting drives: explicit fractal limit set

A single boost yields an orbit (no fractal). A *semigroup* of *non-commuting* boundary-fixing Möbius maps produces a Cantor-like limit set on the boundary.

Concrete generators in $\text{PSL}(2, \mathbb{R})$

Let

$$B(\tau) = \begin{pmatrix} \cosh \tau & \sinh \tau \\ \sinh \tau & \cosh \tau \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad T_a = B(\tau_a), \quad T_b = S B(\tau_b) S^{-1},$$

with $\tau_a \neq \tau_b > 0$. T_a fixes ± 1 ; T_b fixes $S(\pm 1) = \frac{1 \pm 1}{1} = \{0, \infty\}$ (a different axis). Then T_a, T_b are hyperbolic and do not commute; the semigroup $\langle T_a, T_b \rangle$ is Schottky for suitable parameters, and its boundary limit set is fractal.

Proposition 4 (Multi-drive fractality). *For non-commuting T_a, T_b as above, the limit set Λ of $\langle T_a, T_b \rangle$ on $\partial(-1, 1)$ is totally disconnected and of Hausdorff dimension $\Delta \in (0, 1)$ determined by the generators. The induced distribution of rapidities over alternating drive words exhibits self-similarity at scale.*

Remark 2 (Numerical estimator). Generate all reduced words up to length L in $\{T_a, T_b\}$ and record their action on a seed $x_0 \in (-1, 1)$. Project to the boundary by $r = \text{artanh}(x)$ and perform box-counting on $\{r_w\}$; extrapolate Δ as $L \rightarrow \infty$.

8 Engineering falsifier: FLOW-GATE

Signal path

EEG $\rightarrow$ entropy rate Φ (CTW or LZ; 1–40 Hz band, 250 Hz sampling). RNG/SPAD bitstream $\rightarrow$ windowed bias δ (lag-1 alternation excess). Compute $r = \text{artanh}(\delta)$. Regress r on Φ :

$$r_t = \beta_0 + \beta_1 \Phi_t + \varepsilon_t, \quad \hat{\lambda} = \beta_1.$$

Windows, estimators, and CIs

- Bit windows: $N_b \in [10^3, 10^4]$ (50% overlap). EEG windows aligned; Φ in bits/s.
- Estimators: CTW (depth 10) or LZ78; both yield consistent entropy-rate estimates on EEG bands.
- Inference: OLS with HAC (Newey–West) standard errors ($\text{lag} = \lfloor 3\sqrt{T} \rfloor$). Bootstrap confirmation (block bootstrap).

Power target

With $\sigma_r^2 = \text{Var}(r_t)$ and drive variance $V_\Phi = \text{Var}(\Phi_t)$, the slope s.e. is $\text{se}(\hat{\lambda}) \approx \sigma_r / \sqrt{T V_\Phi}$. For detection at 5σ of $|\lambda| \geq 10^{-6}$ with typical $\sigma_r \sim 10^{-2}$ and $V_\Phi \sim 10^2$ (bits/s)², one needs $T \gtrsim 10^6$ aligned windows ($\sim$ hour-scale at 10 Hz effective rate with overlap).

Falsifiers

- **Shuffle test:** time-shuffle $\Phi \Rightarrow \hat{\lambda} \rightarrow 0$ within CIs.
- **Observer swap:** change observer/device $\Rightarrow \hat{\lambda}$ changes systematically if coupling exists.
- **Null device:** LED-flicker pseudo-RNG $\Rightarrow \hat{\lambda}$ consistent with 0.

9 Discussion

The stencil, invariants, and rigidity results make the hyperbolic response law unavoidable under mild, checkable hypotheses. Either $\kappa = 0$ (useful null; the stencil still unifies bounded dynamics) or $\kappa \neq 0$ (a universal, weak participatory coupling). Multi-drive experiments test the fractal prediction directly. FLOW-GATE provides a compact, falsifiable artifact.

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A Energy–rapidity inequality: derivation

Consider a two-outcome channel with baseline Bernoulli parameter $p = \frac{1+u}{2}$ and rapidity $r = \text{artanh}(u) = \frac{1}{2} \log \frac{p}{1-p}$. Changing the distribution from p to p' at temperature T costs at least

$$Q \geq k_B T D_{\text{KL}}(\text{Bern}(p') \parallel \text{Bern}(p)) = k_B T \left[p' \log \frac{p'}{p} + (1-p') \log \frac{1-p'}{1-p} \right].$$

For small increments along a protocol that monotonically tilts log-odds, write $\theta = 2r$ and $d\theta$ the control. The Fisher information in θ is $I_F(\theta) = p(1-p)$, so locally $D_{\text{KL}} \approx \frac{1}{2} I_F (d\theta)^2 = 2p(1-p) (dr)^2$. For a quasi-static path that changes r by Δr while keeping $p(1-p)$ near its baseline average $\bar{q}$, the integrated cost obeys

$$Q \gtrsim k_B T \int 2\bar{q} (dr)^2 \geq \frac{k_B T}{\chi} \Delta r,$$

by Cauchy–Schwarz, where χ^{-1} is the path-dependent factor $\chi^{-1} := 2\bar{q} \Delta r / \int |dr|$. Optimizing over monotone tilts yields a finite device constant χ determined by the admissible actuation schedule and baseline occupancy (bounded away from 0, 1). This proves Proposition 3.

B Worked micro-examples

Ising ($J=0$)

$m = \tanh(\beta h) \Rightarrow r = \operatorname{artanh}(m) = \beta h$ (slope = 1). For weak $J > 0$, linear response in r bends by $O(J)$ via $m = \tanh(\beta J m + \beta h)$.

Logistic learner

$s(t) = 1/(1 + e^{-(\lambda t + b)})$, $u = 2s - 1$, $r = \operatorname{artanh}(u) = \frac{1}{2}(\lambda t + b)$ (exact).

Wright–Fisher

For small selection s , $\dot{p} \approx s p(1 - p)$, hence $\frac{d}{dt} \operatorname{artanh}(2p - 1) = s$.

SIR (phase)

In a single-phase, well-mixed regime, (i) is approximately affine in contact opportunities; remap to $u \in (-1, 1)$ and apply $r = \operatorname{artanh}(u)$.

C References

References

- [1] J. Aczél, *Lectures on Functional Equations and Their Applications*, Academic Press, 1966.
- [2] G. Birkhoff, “Extensions of Jentzsch’s theorem,” *Trans. AMS* **85** (1957), 219–227. (Hilbert/Thompson metric and contraction.)
- [3] M. A. Akian, S. Gaubert, C. Walsh, “The Hilbert metric, contractive mappings, and projective geometry,” various surveys.
- [4] A. A. Ungar, *Analytic Hyperbolic Geometry and Albert Einstein’s Special Theory of Relativity*, World Scientific, 2008.
- [5] S.-I. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS, 2000.
- [6] N. N. Chentsov, *Statistical Decision Rules and Optimal Inference*, AMS, 1982. (Monotone metrics uniqueness.)
- [7] E. Cartan, “Sur les domaines bornés homogènes de l’espace de n variables complexes,” *Abh. Math. Sem. Univ. Hamburg* **11** (1935), 116–162.
- [8] M. Koecher, “The Minnesota notes on Jordan algebras and their applications,” (1962); E. B. Vinberg, “Homogeneous cones,” *Soviet Math. Dokl.* **1** (1960), 787–790.
- [9] D. Sullivan, “Entropy, Hausdorff measures old and new, and limit sets of geometrically finite Kleinian groups,” *Acta Math.* **153** (1984), 259–277.
- [10] R. Bowen, “Hausdorff dimension of quasicircles,” *Publ. Math. IHES* **50** (1979), 11–25.
- [11] R. Landauer, “Irreversibility and heat generation in the computing process,” *IBM J. Res. Dev.* **5** (1961), 183–191. (Information thermodynamics.)